Malachi Mabie

Robotics Software and Machine Learning Engineer

Carpinteria, CA | Open to relocation to Los Angeles South Bay and Orange County

malachimabie.com | linkedin.com/in/mlmabie | github.com/mlmabie

Summary

Robotics software engineer working across motion planning, operator controls, runtime integration, and agent-assisted evaluation. Combines applied machine learning with medical-device manufacturing experience, including robotic tooling from CAD and prototyping through qualification and production use.

Experience

Advanced Metal Research / Gradient Industrial Robotics | July 2026 to Present

Robotics ML Research Engineer (contract)

- Build agent-assisted robotics software spanning motion-planning evaluation, typed task records, candidate trajectories, independent checks, and replayable experiment evidence.
- Develop Steam Deck operator controls and robot runtime integration. Resolved packet, identity, and configured-axis mismatches, contributing to operator-confirmed jogging on the assembled robot.
- Tie evaluation results to code revisions and robot frames, with independent kinematic, collision, calibration, and interface checks. Keep jogging, full trajectory execution, and welding validation separate.

Mabie Industries, LLC | April 2025 to Present

AI Software Engineer / Independent Contractor

- Built and shipped six production systems for Drone Institute, including thermal imagery and GIS workflows, a marketplace, partner onboarding, operations, and order fulfillment. Owned architecture through deployment and production debugging.
- Built agent research infrastructure with typed tools, retrieval, memory, execution traces, replay, and independent verifiers. Delivered cognitive-security threat modeling and evaluation for Siethar Group.

RAM Laboratories, Inc. | April 2022 to March 2025

Senior Member of Technical Staff / Principal Investigator, AI/ML R&D

- Led T-UEBA, a $250,000 Army SBIR for tactical user and entity behavior analytics. Reported program evaluation results of 0.887 F1, under 3% false positives, and 15.2 ms per graph with a 15 MB temporal graph model.
- Directed VULCAN for the DARPA AI Cyber Challenge, combining code models with static analysis, fuzzing, and runtime verification to evaluate generated patches.
- Invented DL-Patcher for automated C/C++ vulnerability repair, with synthetic bug insertion, training-data filtering, checker feedback, and human review. U.S. patent application 18/375,839 filed.

Applied Medical | June 2013 to September 2018

Polymer Design and Mechanical Engineer; previously Process Development Intern

- Designed and qualified robotic end-of-arm tooling, pneumatic mechanisms, and manufacturing fixtures. A production insert-molding process fix reduced operating variance by approximately $100,000 per month.

Selected project

DXF geometry workbench: recovered 2,237 polygons from approximately 67,000 CAD primitives across 111 layers, with parameter sweeps and SVG review artifacts. github.com/mlmabie/DXF_Primitive_to_Polygon

Education

University of California San Diego | B.S. Cognitive Science, 2021

Specialization in Machine Learning and Neural Computation.

Technical skills

Python, C/C++, PyTorch, TypeScript, SQL; motion planning, CAD geometry, robot interfaces, collision and kinematic checks, evaluation, GIS, computer vision; AWS, Docker, Linux.